UNIT – IV

PROGRAMMABLE LOGIC DEVICES

Programmable Logic Devices (PLDs)

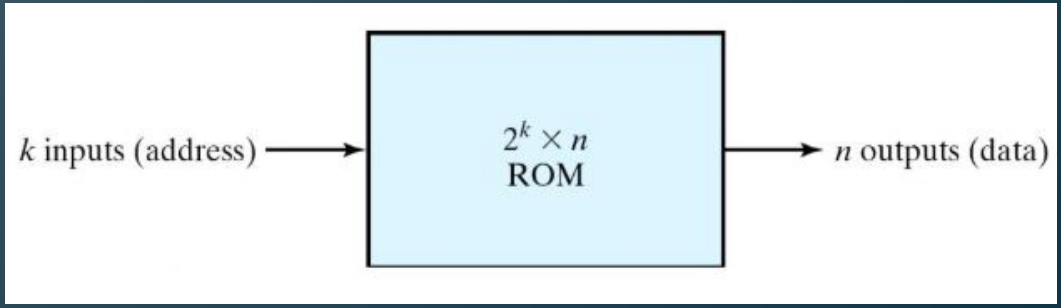
A programmable logic device is defined as an integrated circuit that contains an array of logic elements and interconnections that can be programmed by the user to implement a desired logic function. The logic elements are usually simple combinational or sequential circuits, such as AND, OR, NOT, and XOR gates, or registers. The interconnections are usually programmable switches or multiplexers that can connect the inputs and outputs of the logic elements in different ways.

Memory

- It is a device to which binary information is transferred for storage and from which information is retrieved when needed for processing.
- When data processing takes place, information from memory is transferred to selected registers in the processing unit.
- Thus, a “Memory unit” is a collection of cells capable of storing a large quantity of binary information.
- There are two types of memories: Random Access Memory (RAM) and Read Only Memory (ROM).
- RAM stores new information. RAM can perform both write and read operations.
- ROM can perform only the read operation.

Read Only Memory (ROM)

- A ROM is a memory device in which permanent binary information is stored.
- As the name suggests it is meant only for reading the information from it.
- ROMs can be used for designing combinational logic circuits.
- ROM consists of ‘k’ inputs and ‘n’ outputs.
- The inputs provide the address for memory and the outputs give the data bits of the stored word that is selected by the address. The number of words in a ROM is determined from the fact that ‘k’ address input lines are needed to specify ‘2^k’ words.



Types of ROMS

- The process of entering the information into a ROM is referred to as *programming* the ROM. Depending upon the programming process employed, the ROMs are classified as:
- 1. *Mask Programmable Read Only Memory (MROM):*
- In this type of ROM, the user specifies the data to be stored to the manufacturer of the memory. Once programmed, the data pattern can never be changed.
- 2. *Programmable Read Only Memory (PROM):*
- This type of memory comes from the manufacturer without any data stored in it i.e., empty. The data pattern is programmed electrically by the user. Once programmed, the data cannot be altered.

Types of ROMS (Contd.)

- 3. *Erasable Programmable Read Only Memory (EPROM):*
- In this type of memory, data can be written any number of time i.e., they are reprogrammable. Before it is reprogrammed, the content already stored is erased by exposing the chip to UV radiation for about 30 minutes.
- 4. *Electrically Erasable and Programmable Read Only Memory (EEPROM):*
- This is another type of reprogrammable memory in which erasing is done electrically rather than exposing the chip to the UV radiation.

Programmable Logic Devices (PLDs)

- A Programmable Logic Device is an IC that is user configurable and is capable of implementing logic functions.
- A PLD contains a large no. of gates, flip-flops, and registers that are interconnected on the chip.
- They contain an array of AND gates and another array of OR gates.
- There are three kinds of PLDs based on the type of arrays, which has programmable feature.
  - Programmable Read Only Memory (PROM)
  - Programmable Array Logic (PAL)
  - Programmable Logic Array (PLA)

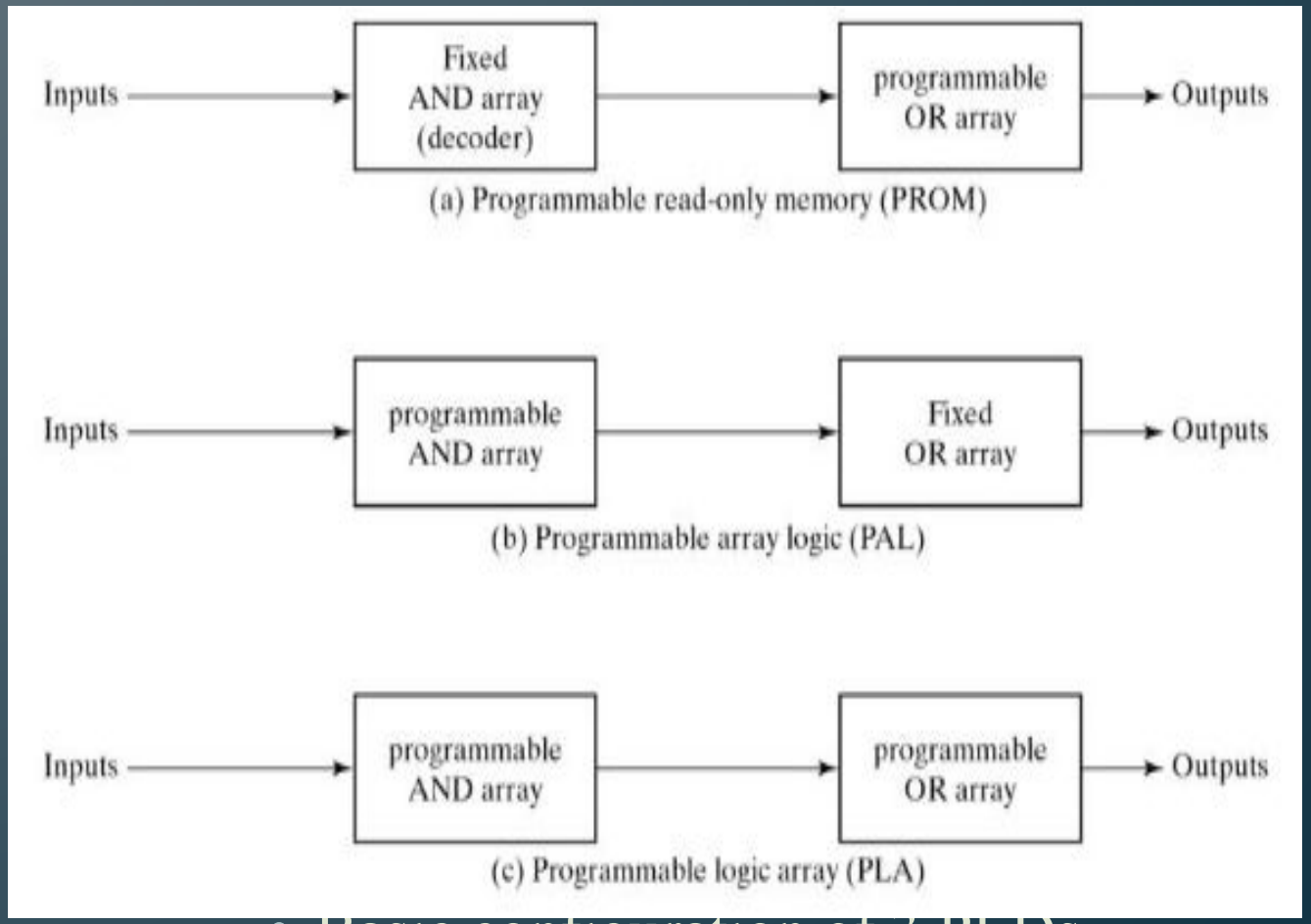
Programmable Logic Devices (PLDs)

- The process of entering the information into these devices is known as **Programming**.
- Basically, users can program these devices or ICs electrically in order to implement the Boolean functions based on the requirement.
- Here, the term programming refers to hardware programming but not software programming.
- Advantages of PLDs over fixed function ICs:
  - Low cost
  - Low power
  - Easy testing
  - Less design time
  - Low space
  - High reliability
  - Easy design
  - High speed

Combinational PLDs

- Combinational PLD is an IC with programmable gates divided into AND & OR array to provide an AND-OR sum-of-product implementation. There are 3 major types of combinational PLDs:
- 1. PROM has a fixed AND array and a programmable OR array. The programmable OR gates implement the Boolean functions in sum-of-minterms form.
- 2. PAL has a programmable AND array and a fixed OR array. The AND gates are programmed to provide the product terms for the Boolean functions, which are logically summed in each OR gate.
- 3. The most flexible PLD is the PLA in which both the AND array and OR arrays can be programmed. The product terms in the AND array may be shared by any OR gate to provide the required sum-of-products implementation.

Combinational PLDs (Contd.)

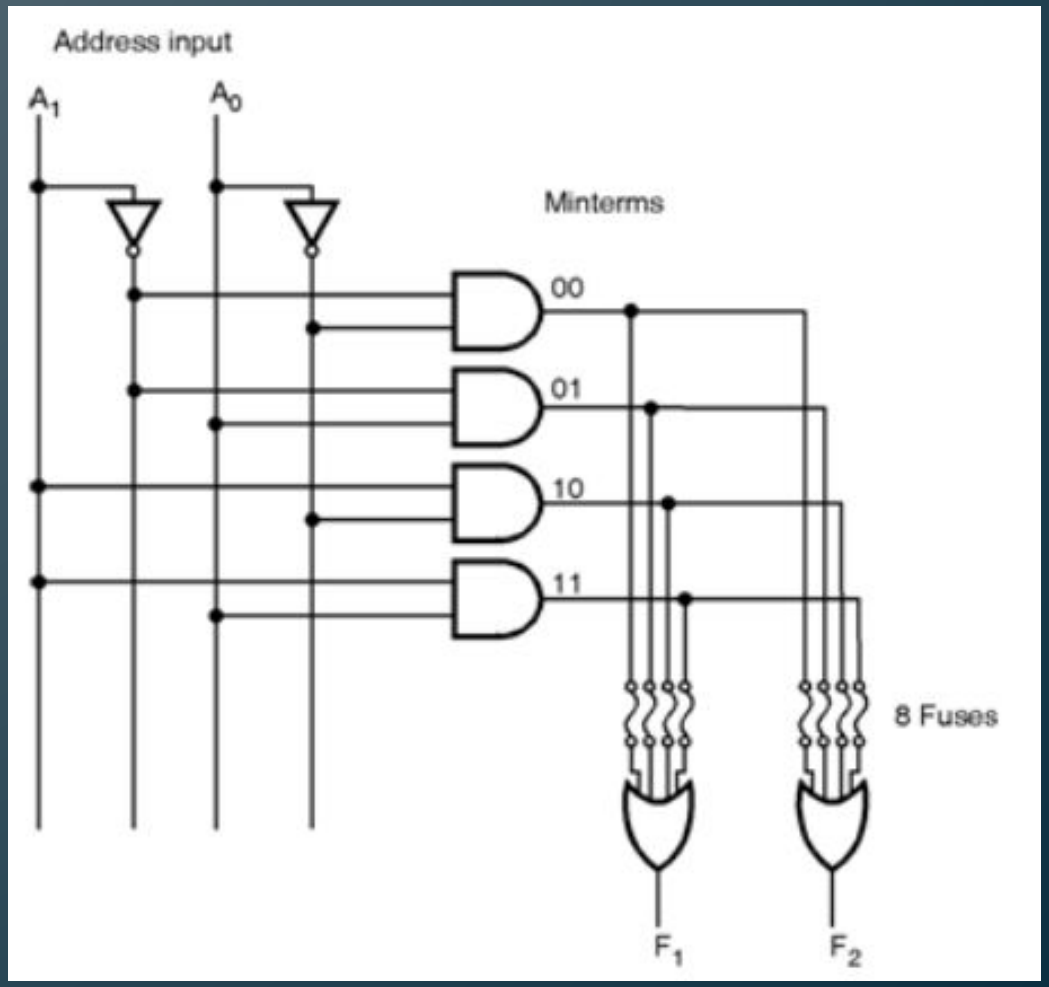


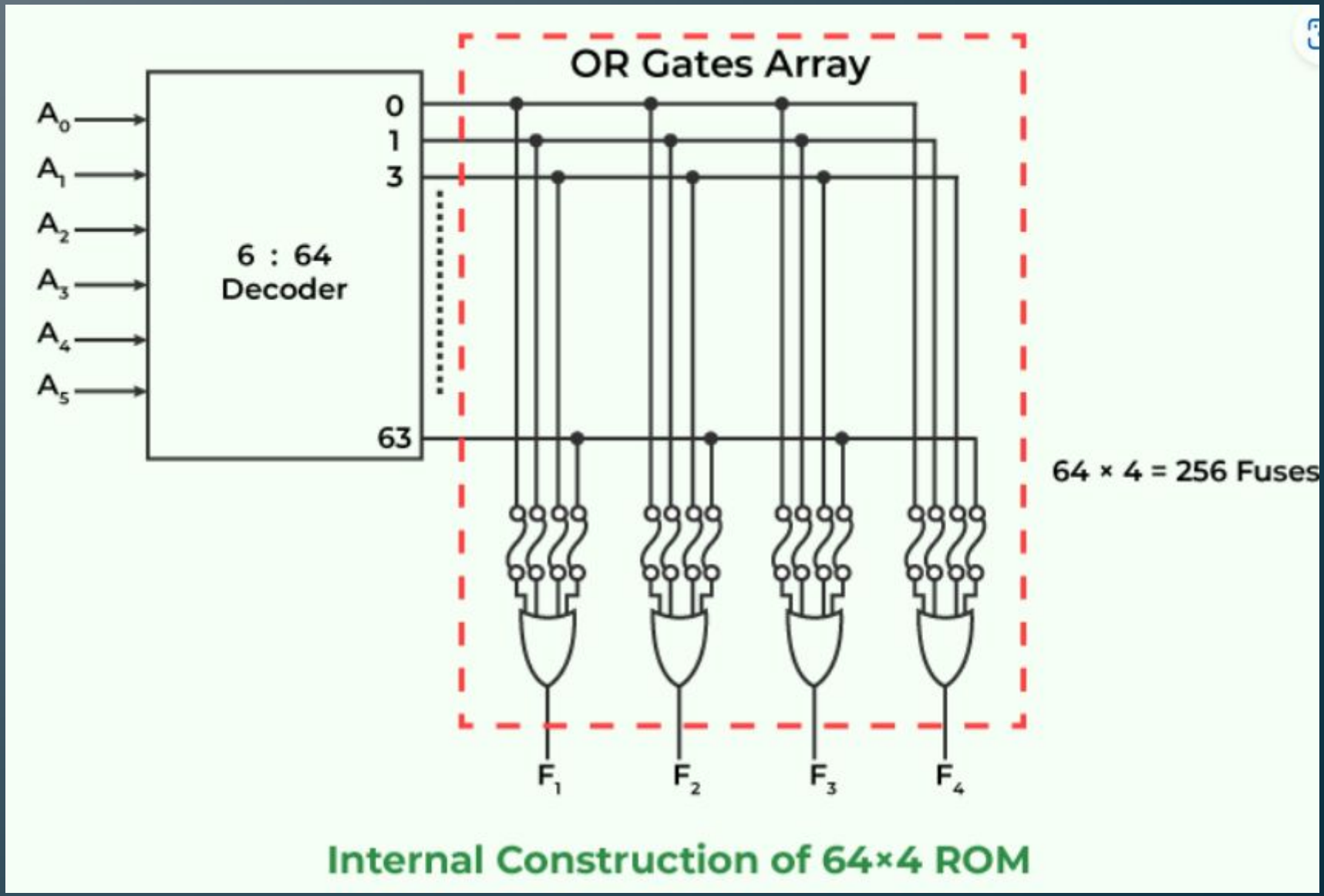
- Basic configuration of 3 PLDs

Programmable Read Only Memory (PROM)

ROM is non-volatile nature memory that means its data will remain unchanged whenever turned off power supply.

2X4 ROM

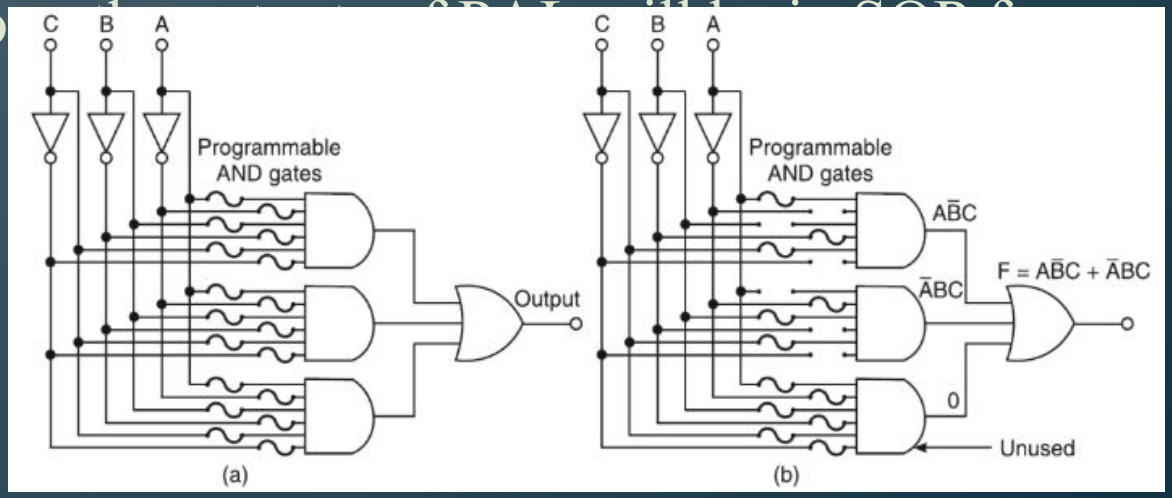




Internal Construction of 64×4 ROM

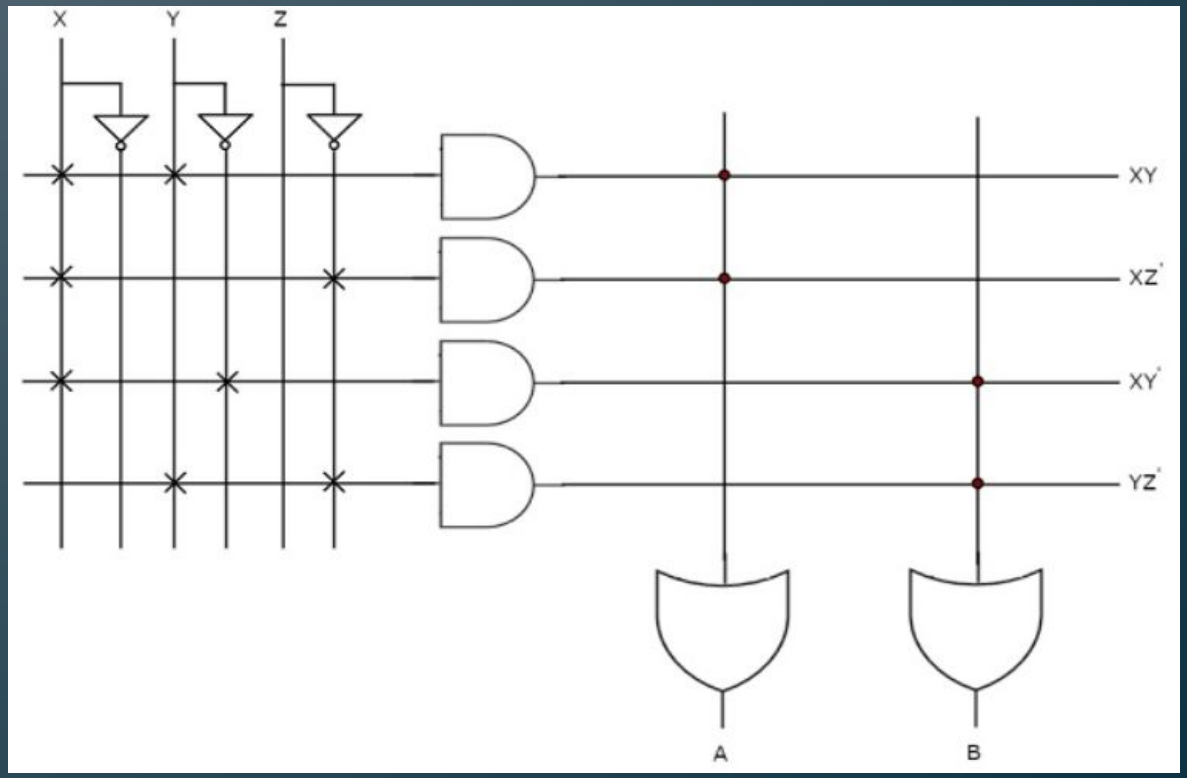
Programmable Array Logic (PAL)

- Inputs of AND gates are programmable i.e., each AND gate has both normal & complemented inputs. Based on requirement, we can program any of those inputs. Thus the required product terms by using these AND gates are generated.
- Inputs of OR gates are not of programmable type. So, the number of inputs to each OR gate will be of fixed type. Hence, apply those required product terms to each OR gate as inputs. Therefo



PAL Example

- Implementing the following Boolean functions using PAL.
  - A=XY+XZ′ B=XY′+YZ′
- The symbol ‘x’ is used for programmable connections.
- The symbol ‘•’ is used for fixed connections.



PAL Programming Table

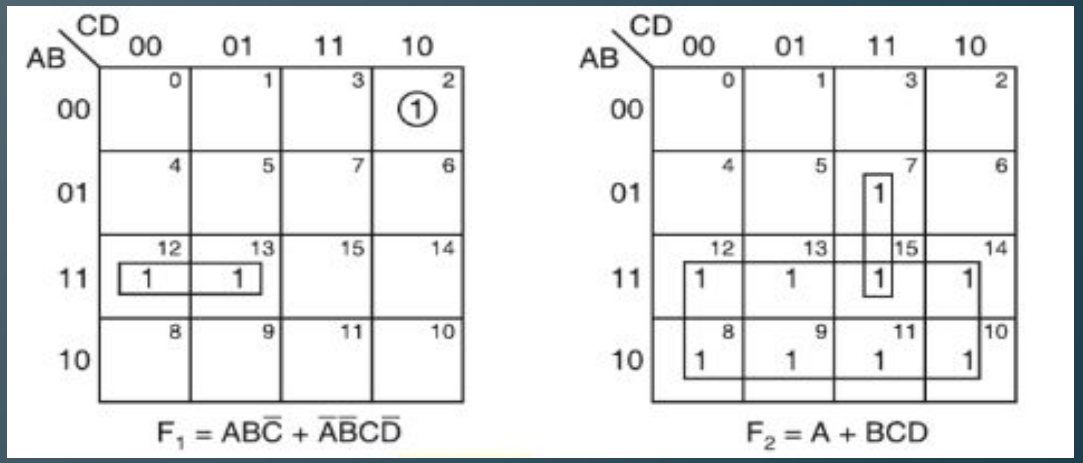
- The PAL programming table consists of 3 columns.
- The first column lists the product terms numerically.
- The second column specifies the required paths between inputs and AND gates.
- The third column specifies the outputs of the OR gates.
- For each product term, inputs are marked with 1, 0, or – (dash).
- If a variable in the product term appears in its true form, the corresponding input variable is marked with a 1.
- If it appears in complemented form, the corresponding input variable is marked with a 0.
- If the variable is absent in the product term, it is marked as a – (dash).

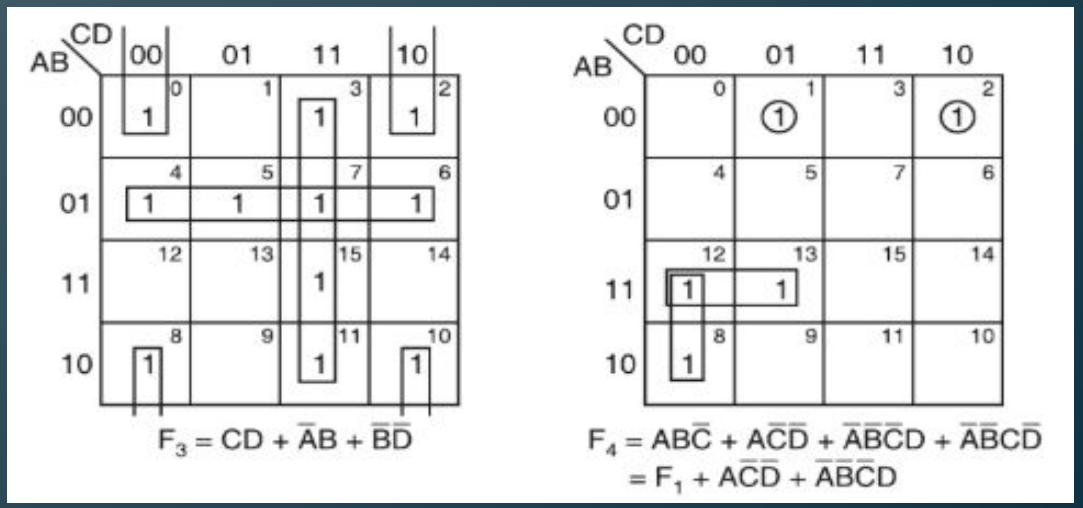
Example 1

- Implement the following Boolean functions using PAL with four inputs and 3-wide AND-OR structure. Also write the PAL programming table.
- $F_1(A, B, C, D) = \Sigma\, m(2, 12, 13)$
- $F_2(A, B, C, D) = \Sigma\, m(7, 8, 9, 10, 11, 12, 13, 14, 15)$
- $F_3(A, B, C, D) = \Sigma\, m(0, 2\text{-}8, 10, 11, 15)$
- $F_4(A, B, C, D) = \Sigma\, m(1, 2, 8, 12, 13)$

Example 1 - Solution

- K-maps for before expressions





Example 1 - Solution

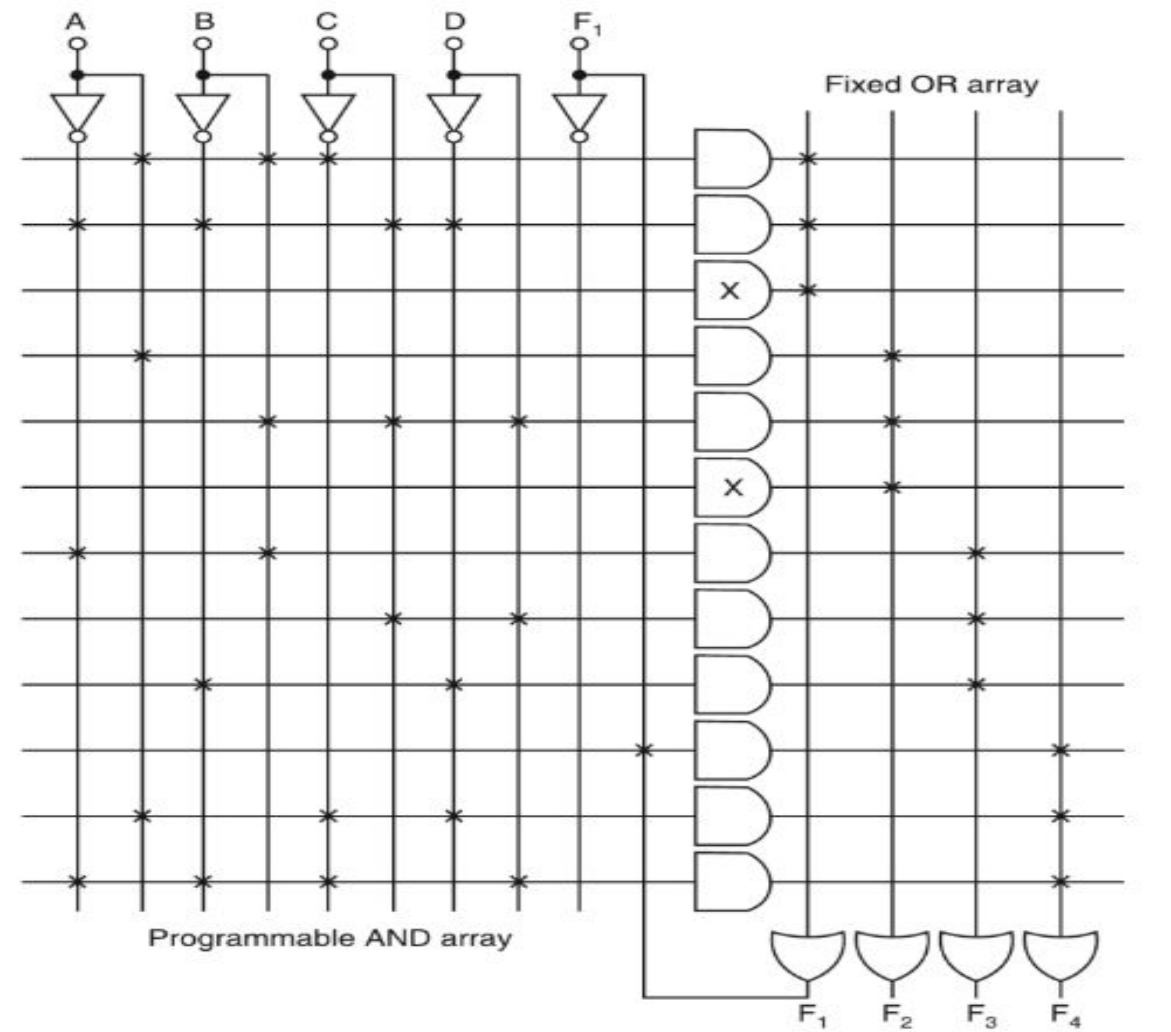
- PAL programming table

Product term	AND Inputs					Outputs
	A	B	C	D	F_1	
1	1	1	0	–	–	$F_1 = AB\overline{C} + \overline{A}\overline{B}C\overline{D}$
2	0	0	1	0	–	
3	–	–	–	–	–	
4	1	–	–	–	–	
5	–	1	1	1	–	$F_2 = A + BCD$
6	–	–	–	–	–	
7	0	1	–	–	–	$F_3 = \overline{A}B + CD + \overline{B}\overline{D}$
8	–	–	1	1	–	
9	–	0	–	0	–	
10	–	–	–	–	1	$F_4 = F_1 + A\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D$
11	1	–	0	0	–	
12	0	0	0	1	–	

(a) PAL programming table

Example 1 - Solution

- Realization of the function using PAL



(b) Realization of the example functions using PAL

Example 2

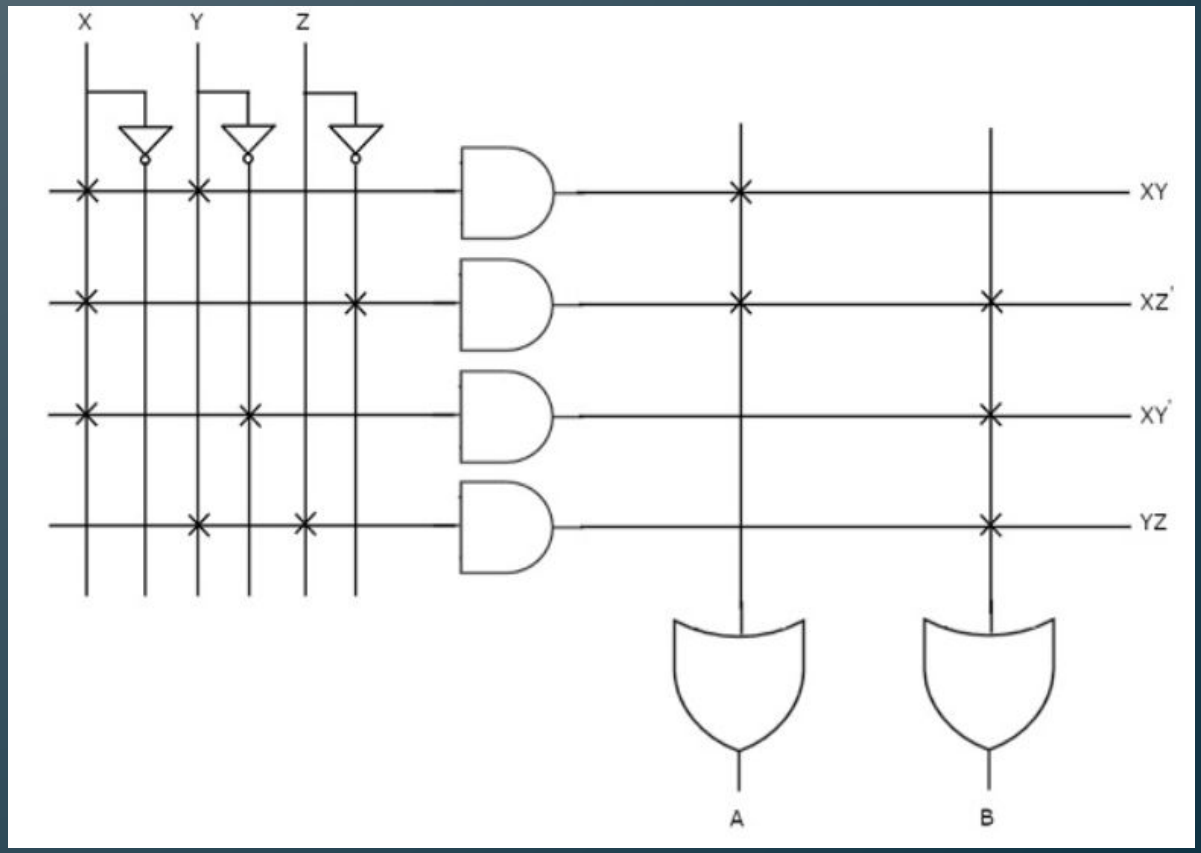
- Implement the following Boolean functions using PAL with four inputs and 3-wide AND-OR structure. Also write the PAL programming table.
- $F_1(A, B, C, D) = \Sigma\, m(6, 8, 9, 12\text{-}15)$
- $F_2(A, B, C, D) = \Sigma\, m(1, 4\text{-}7, 10\text{-}13)$
- $F_3(A, B, C, D) = \Sigma\, m(4\text{-}7, 10, 11)$
- $F_4(A, B, C, D) = \Sigma\, m(4\text{-}7, 9\text{-}15)$

Programmable Logic Array (PLA)

- Inputs of AND gates are programmable. That means each AND gate has both normal and complemented inputs of variables. So, based on the requirement, we can program any of those inputs. So, we can generate only the required product terms by using these AND gates.
- Inputs of OR gates are also programmable. So, we can program any number of required product terms, since all the outputs of AND gates are applied as inputs to each OR gate. Therefore, the outputs of PAL will be in SOP form.

PLA Example

- Implement the following Boolean functions using PLA.
  - A=XY+XZ′ B=XY′+YZ+XZ′

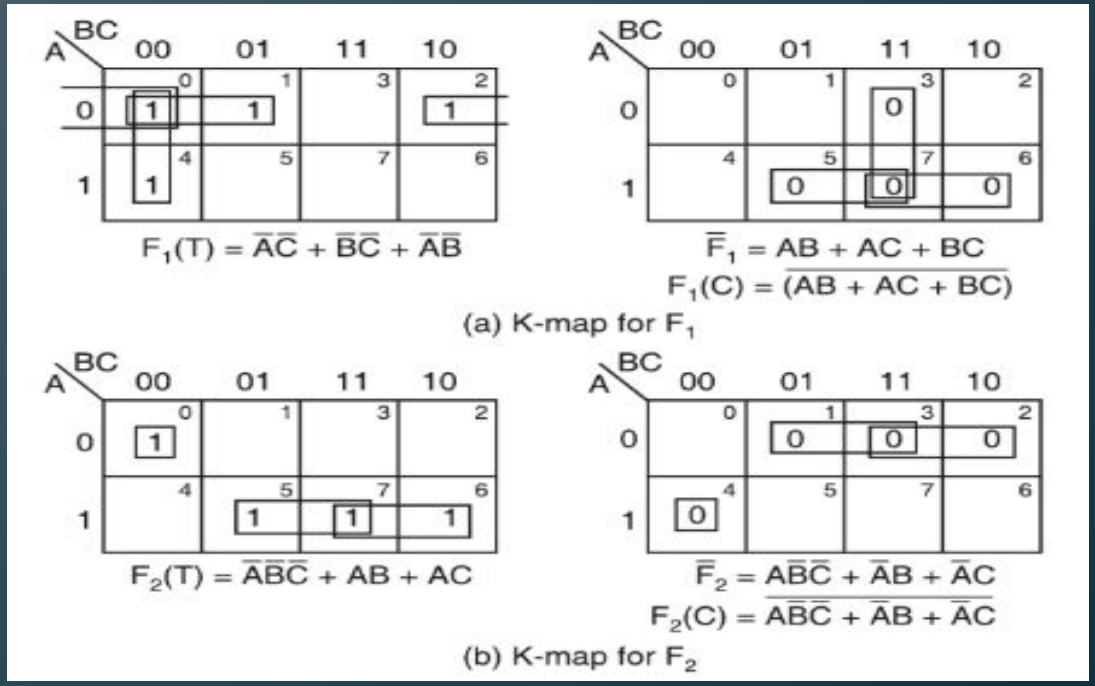


PLA Programming Table

- The PLA programming table consists of 3 columns.
- The first column lists the product terms numerically.
- The second column specifies the required paths between inputs and AND gates.
- The third column specifies the paths between AND gates and OR gates.
- For each output variable, we may have a T (for true) or C (for complement) for programming the X-OR gate. The product terms listed on the left are not part of the table; they are included for reference only.
- a T (true) output dictates that the other input of the corresponding X-OR gate be connected to 0, and a C (complement) specifies a connection to 1.

Example 1

- Implement the following two Boolean functions with a PLA
- $F_1(A, B, C) = \Sigma\, m(0, 1, 2, 4)$
- $F_2(A, B, C) = \Sigma\, m(0, 5, 6, 7)$
- Solution:
- K-maps for above expressions:



Example 1 - Solution

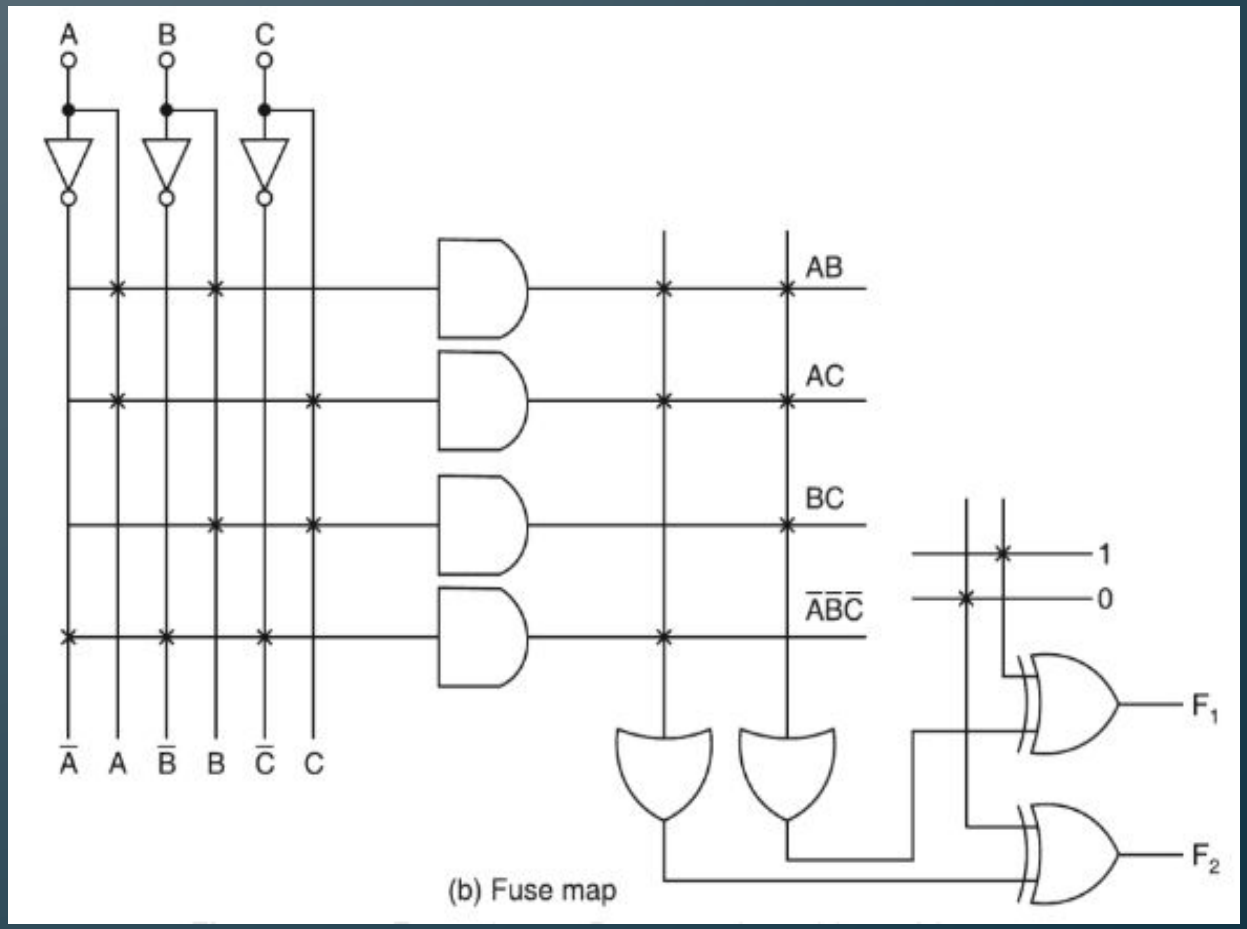
- PLA programming table

Product term		Inputs			Outputs	
		A	B	C	(C) F_1	(T) F_2
1	AB	1	1	–	1	1
2	AC	1	–	1	1	1
3	BC	–	1	1	1	–
4	$\overline{A}\overline{B}\overline{C}$	0	0	0	–	1

(a) PLA programming table

Example 1 - Solution

- Realization of the function using PLA



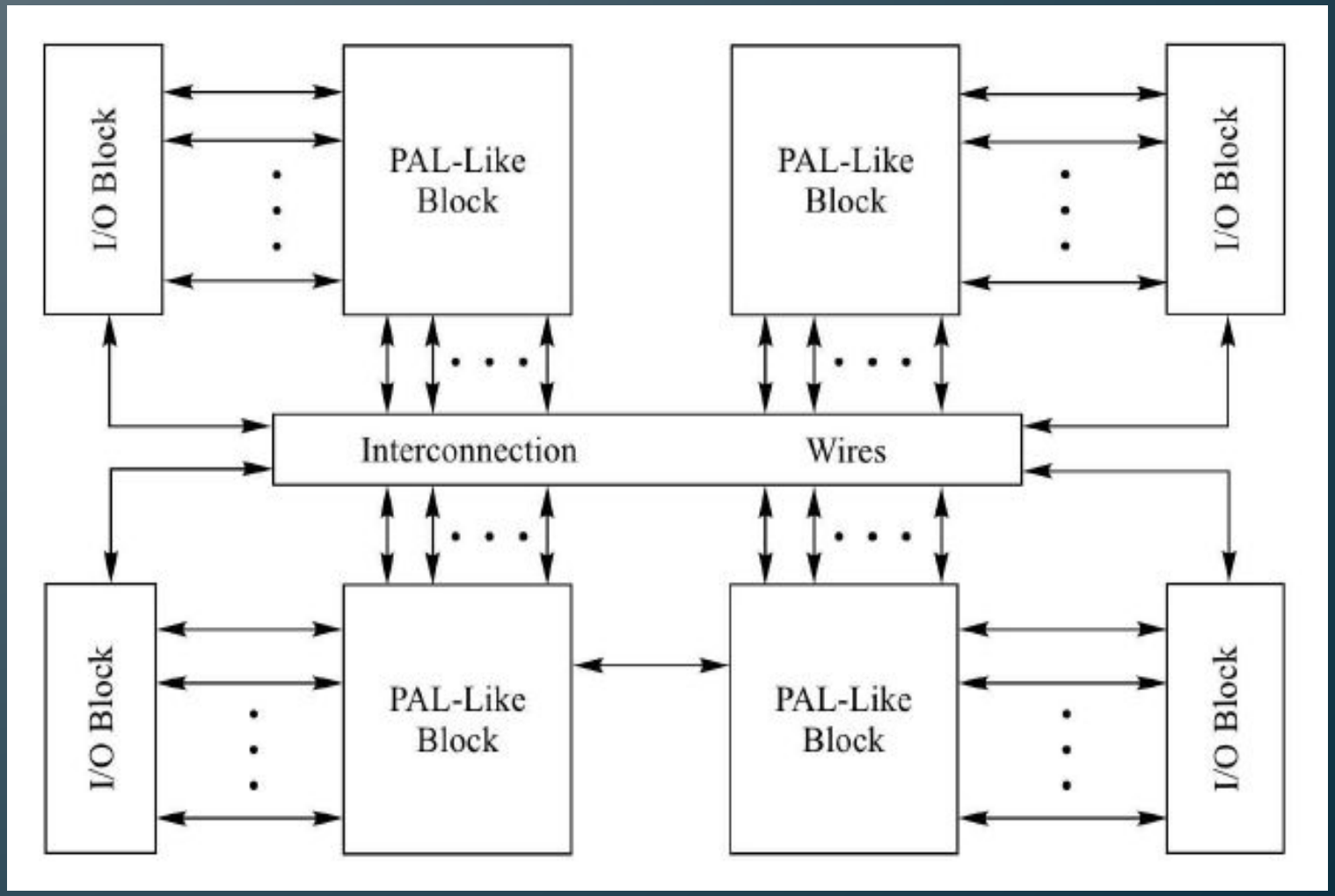
Comparison of PLA and PAL

S.No	Key Point	PLA	PAL
1.	Stands for	Programmable Logic Array	Programmable Array Logic
2.	Construction	Programmable array of AND gates and OR gates.	Programmable array of AND gates and fixed array of OR gates.
3.	Availability	Less prolific	More readily available
4.	Flexibility	Provides more programming flexibility.	Offers less flexibility, but more likely used.
5.	Cost	Expensive	Intermediate cost
6.	Number of functions	Large number of functions can be implemented.	Provides the limited number of functions.
7.	Speed	Slow	High
8.	Complexity	High	Low
9.	Usage	Less used than PAL	More used than PLA

Complex Programmable Logic Device (CPLD)

- The Simple PLDs (SPLDs) such as PALs etc., have limited number of inputs, product terms, and outputs.
- These devices, therefore, can support upto 32 total no. of inputs and outputs only.
- For implementation of circuits that require more inputs and outputs can be employed using more sophisticated type of chip referred to as Complex Programmable Logic Device (CPLD).

Architecture of CPLD

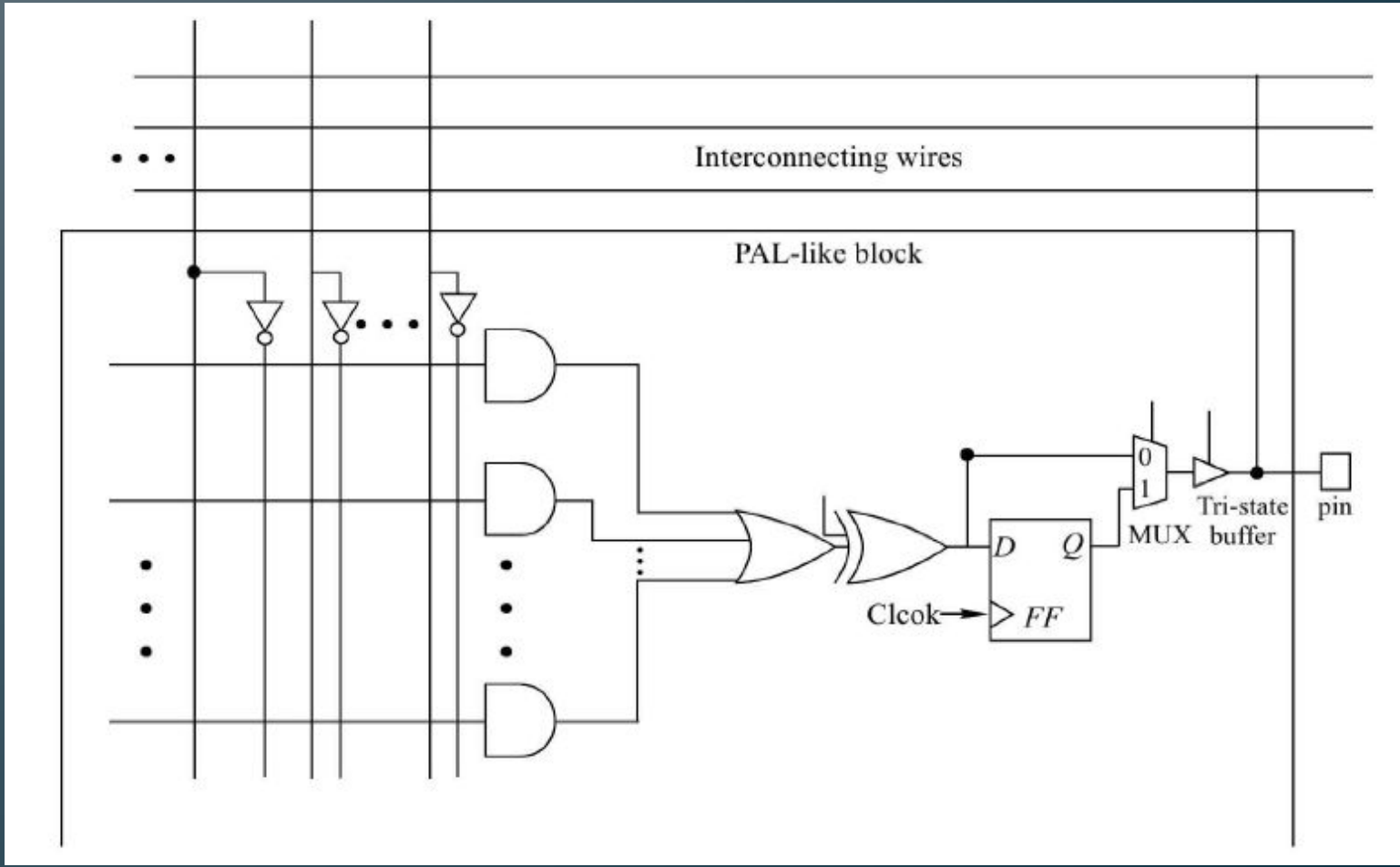


Architecture of CPLD

- A PAL-like block usually consists of about 16 macrocells.
- Each macrocell consists of an AND-OR configuration, an EX-OR gate, a FLIP-FLOP, a multiplexer, and a tri-state buffer.
- Each AND-OR configuration usually consists of 5-20 AND gates and an OR gate with 5-20 inputs.
- An EX-OR gate is used to obtain the output of OR gate in inverted or non-inverted form depending upon its other input being 1 or 0 respectively.
- A D-FF stores the output of the EX-OR gate, a multiplexer selects either the output of the D-FF or the output of the EX-OR gate depending upon its select input 1 or 0.
- The tri-state buffer acts as a switch which enables the chip's pin to be used either as an output (tri-state enabled) or as an input (tri-state disabled).

Architecture of CPLD

- A typical macrocell of a CPLD

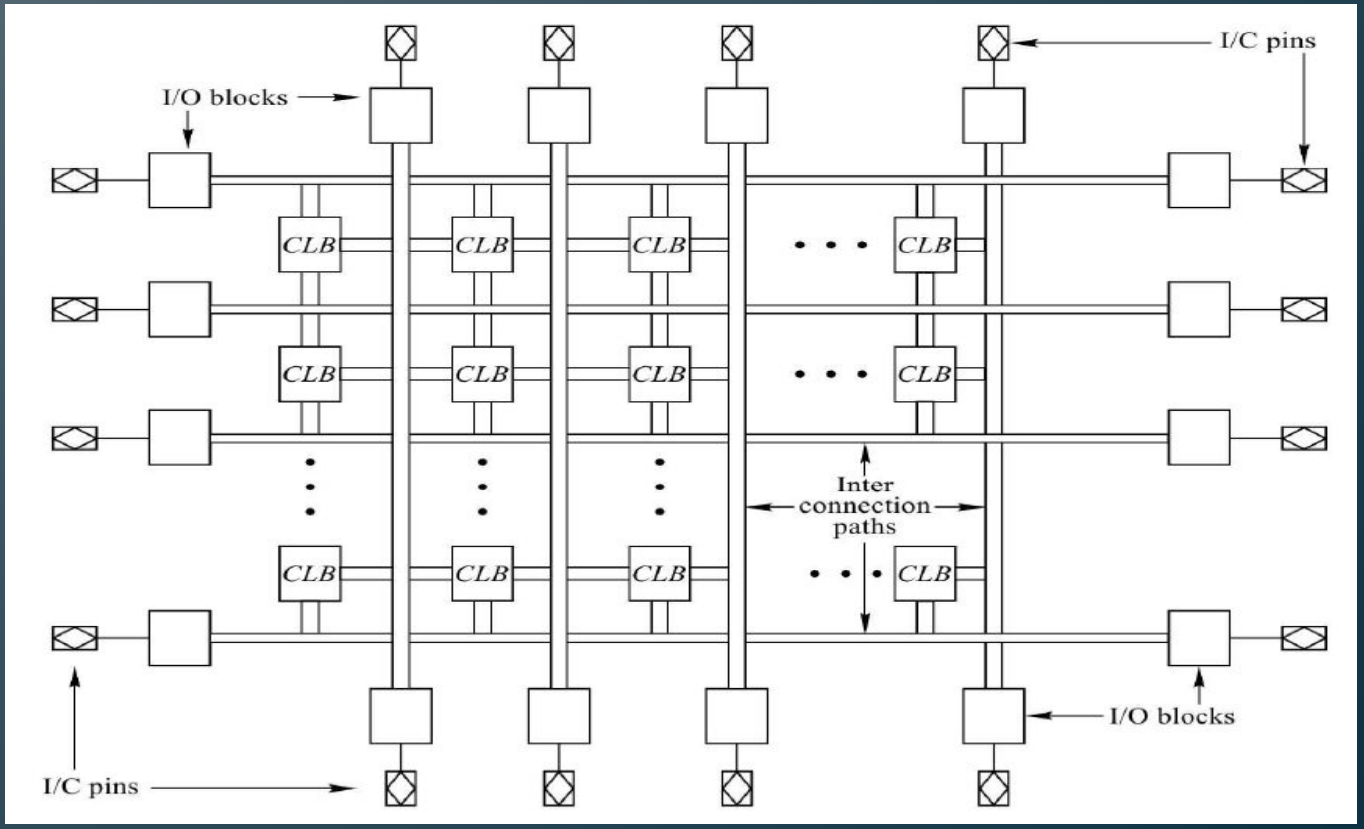


Field Programmable Gate Array (FPGA)

- FPGA contains ten thousand to more than a million logic gates with programmable interconnection.
- The logic densities of FPGAs are much higher than those of CPLDs.
- An FPGA is composed of a no. of relatively independent configurable logic blocks (CLBs), configurable I/O blocks, and programmable interconnection paths.
- All the resources of the device are uncommitted and that these must be selected, configured and interconnected by a user to form a logic circuit for application.

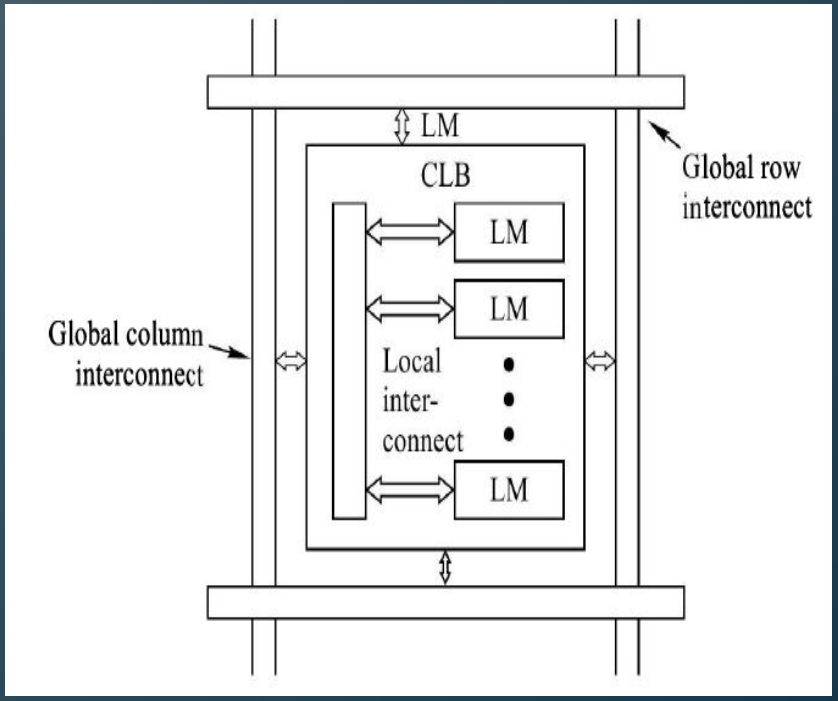
Architecture of FPGA

- The basic FPGA architecture consists of an array of Configurable Logic Blocks (CLBs). The logic blocks are surrounded by configurable I/O blocks. There are rows and columns of programmable interconnection paths.



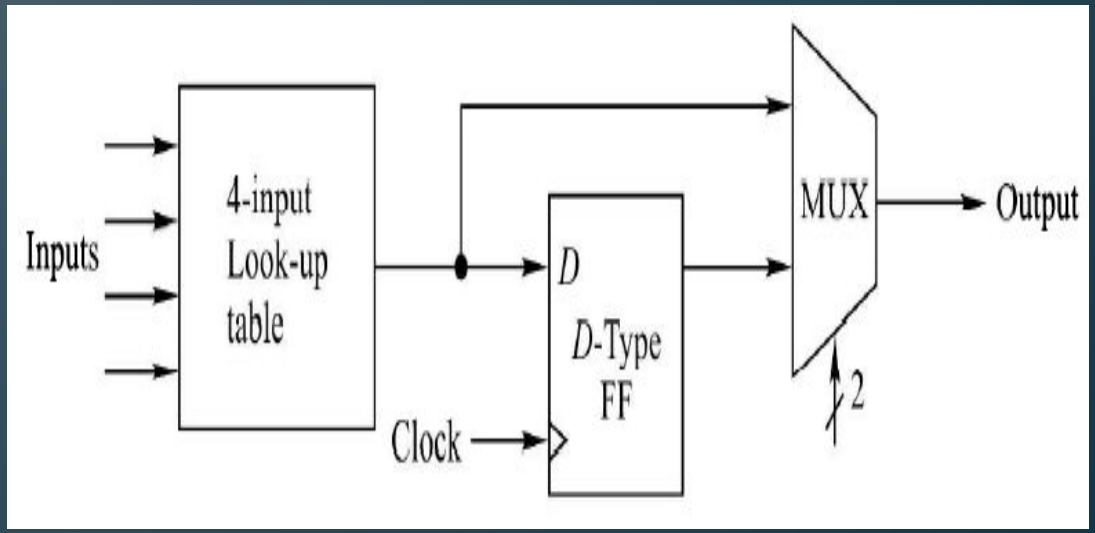
FPGA - CLBs

- A logic block consists of a no. of logic modules (LMs).
- The logic modules are the basic logic elements in an FPGA.
- The logic modules within a CLB are connected through local programmable interconnects.



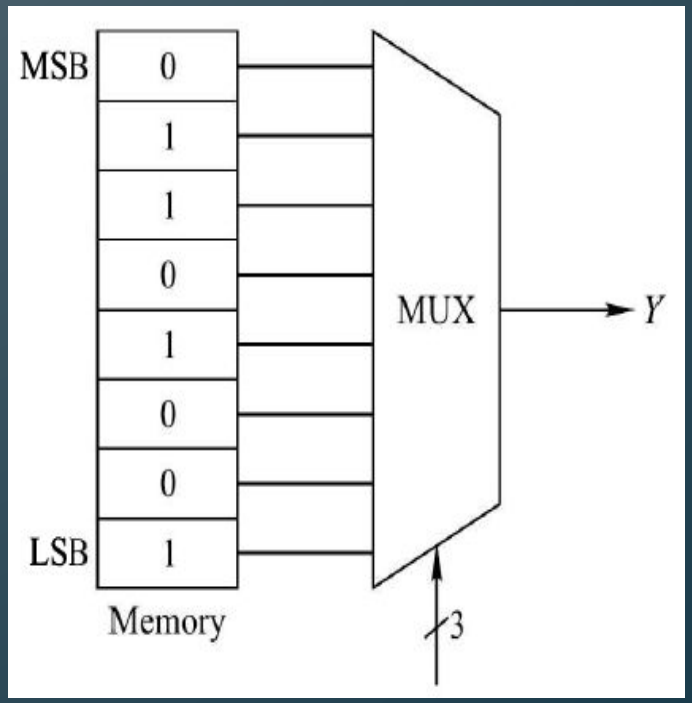
FPGA - LMs

- A logic module consists of an LUT (look-up table), a D-type flip-flop and a multiplexer (MUX).
- The output of the LUT becomes the output of the LM either directly or through D-type flip-flop.
- Thus, the output can be configured for combinational or registered i.e., through flip-flop.



FPGA – LUT

- An LUT consists of a programmable memory and it can be used to generate logic function in SOP form.
- LUT consists of a memory and a MUX.
- If we use an 8-bit memory, an 8:1 multiplexer is required.



FPGA Programming Technologies

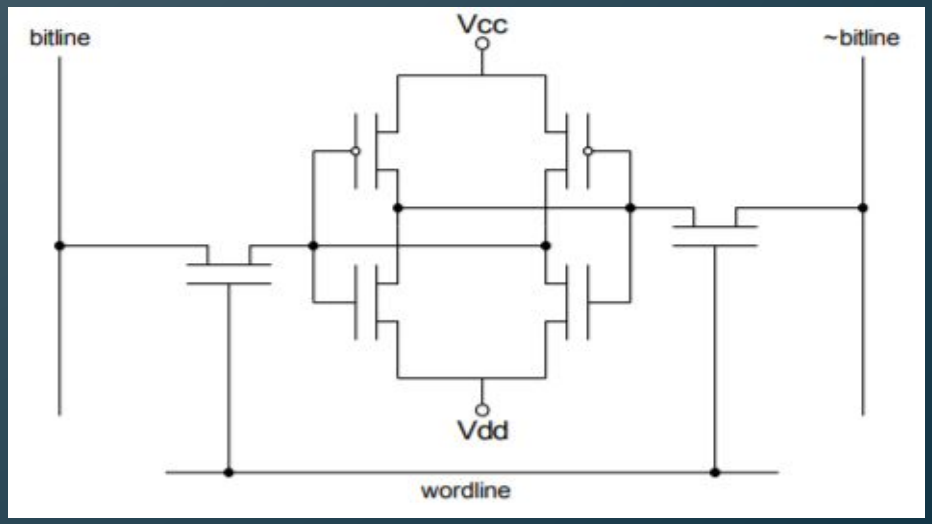
- Following are the FPGA programming technologies:
- SRAM programming technology
- Antifuse programming technology
- EPROM programming technology

SRAM Programming Technology

- SRAM-based FPGA stores logic cells configuration data in the static memory (organized as an array of latches).
- Since SRAM is volatile and can't keep data without power source, such FPGAs must be programmed (configured) upon start.
- There are two basic modes of programming:
- ***Master mode***, when FPGA reads configuration data from an external source, such as an external flash memory chip.
- ***Slave mode***, when FPGA is configured by an external master device, such as a processor. This can be usually done via a dedicated configuration interface or via a boundary-scan interface.

SRAM Programming Technology

- *Advantage:* The major advantage of this technology is that FPGA can be reconfigured (in-circuit) very quickly and can be produced using a standard CMOS process technology.
- *Disadvantage:* The chip area required by SRAM approach is relatively large. This is because at least 5 transistors are needed for each RAM cell, as well as the additional transistors for the pass gates or multiplexers.
    - SRAM memory cell



Antifuse Programming Technology

- The antifuse is a device that doesn't conduct current initially, but can be “burned” to conduct current.
- The antifuse behavior is thus opposite to that of the fuse, hence the name.
- The antifuse-based FPGA can't be then reprogrammed since there is no way to return a burned antifuse into the initial state.

- *Advantage:* Small area overhead.
- *Disadvantage:* Unwanted Long Delay. It is only one-time programmable.

EPROM Programming Technology

- EPROM Cell is almost as small as Antifuse.
- We can use an EPROM transistor as a programmable switch.
- EPROM technology is sometimes called as Floating-Gate Avalanche MOS Technology (FGMOS or FAMOS).

✔ An unprogrammed n-channel EPROM transistor acts as an ON switch for normal operating voltage.

✔ A programmed n-channel EPROM transistor acts as OFF.

- An n-channel EPROM transistor is programmed by applying higher-than-operation voltages to the drain and the control gate.
- Erasable by exposing the floating gate to UV light.

EPROM Programming Technology

- *Advantage:* It requires re-programmable but do not require external storage.
- *Disadvantage:* The resistor consumes static power.

- A section of an EPROM-based device

